

What's New in Xraster Version 04

Installation Notes

Installation of the new **Xraster version 04** must be performed **after uninstalling the previous version**, in accordance with the instructions provided in the **User Manual**.

This version additionally requires the installation of the python **scipy library**.

Changes Compared to Version 03

1. Program Stability Fix

A bug causing the program to close when the **ENTER** key was pressed has been fixed (reopening the program restored the state from before it closed).

In the current version, pressing **ENTER** does **not trigger any action**.

2. DATA Tab Enhancements

A new option “**Flatten variable to bottom**” has been introduced.

This option:

- Flattens four-dimensional data (x, y, z, t) into three-dimensional data (x, y, t) *using* the variable's values at the bottom layer,
 - Creates:
 - a new flattened variable, and
 - an additional variable containing depth information.
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3. SUBSET Tab Improvements

- For data types **3-1** and **4-11** (three-dimensional data with a time dimension), it is now possible to transform a variable into its **anomaly relative to the mean value**, calculated **separately for each grid cell**.
- When a new subset is created by pressing **OK**:
 - Information about the **data type** (code and description) is displayed below the **OK** button.
 - Above the **RESET** button, the following information is shown:
 - the **total area** of all cells (in km², if the km2 variable has been created),
 - the **total number of cells**.

4. New GIS Tab

A new **GIS tab** has been added, performing two independent tasks:

4.1 Time Series Computation for Water Bodies

This tool computes a **time series of mean values** for a selected variable within a specified **water body**.

4.2 Temporal Region Groups

This feature introduces the concept of a **temporal region group**, along with tools for its computation and analysis. It extends existing 2D GIS functions (such as *region group*), which assign unique IDs to spatially contiguous raster cells with the same value. **Temporal region groups** preserve contiguity not only in space, but also **over time**.

Documentation

All changes are described in detail in:

Xraster_QGIS_Plugin_user_manual_EN_04.pdf

A new **Tutorial** covering the changes and newly introduced tools is planned to be ready in **February 2026**.
